JASH THAKKAR
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EDUCATION

Georgia State University – Atlanta, GA
Bachelor of Science in Computer Science | May 2026

- Relevant Coursework: Data Structures, Algorithms, Systems Programming, Operating Systems, Software Engineering, Software Development, Data Science Fundamentals, Calculus I and II, Linear Algebra, Statistics, Mobile Development, Web Programming, Cybersecurity Fundamentals

WORK EXPERIENCE

Freelance
Software Engineer | Georgia | June 2024 – Present

- Developed **6+ full stack web applications** for local businesses using **HTML, CSS, JavaScript, PHP, and SQL** delivering scalable, user-friendly, and production ready software.
- Deployed websites to production, using **Vercel** hosting, **MySQL** for database configuration, **Git** based version control, and domain setup.
- Improved product usability, accessibility, and performance, directly contributing to an average of **20%** increase in user engagement through client projects.
- Gathered requirements through conversations with clients and translated business needs to technical solutions demonstrating strong communication and product thinking skills.
- Implemented authentication systems, including flows for logging in and creating accounts, password hashing, session management, and input validation to prevent security risks.

Select Management Resources
Software Engineer Intern | Alpharetta, Georgia | January 2023 – July 2023

- Developed **Python** and **JavaScript scripts** to streamline troubleshooting workflows, reduce repetitive operational tasks, and improve incident response efficiency.
- Designed and maintained structured technical documentation, system guides, and process workflows, improving onboarding efficiency by **20%.**
- Investigated application, system, and network issues through log analysis, ticket data review, and root cause analysis, contributing to a **25%** reduction in recurring incidents.
- Collaborated with infrastructure and operations teams to identify system reliability gaps and implement tooling and process improvements.
- Debugged and resolved complex software and system issues using structured problem solving, scripting-based validation, and logical decomposition.
- Produced internal data driven reports and dashboards to track issue trends and performance metrics, enabling data driven operational decisions.

PROJECTS

A/B Testing Engine (Python, Streamlit, scikit-learn, Pandas, NumPy, SciPy, Plotly)

- Designed an end-to-end experimentation analytics platform that uses real, and simulated A/B testing data to perform statistical significance testing using two proportion z-tests and t-tests to evaluate features.
- Implemented a hybrid modeling system that combines **Logistic Regression and Random Forest** models to predict user conversion probability, compare model performance and generate a prediction using confidence through **ROC-AUC.**
- Engineered feature processing pipelines for user behavior data, that included data validation, transformation, and aggregation. Allowing for a robust analysis across both uploaded and generated datasets.
- Built a decision engine that uses statistical results and ML predictions to produce a rollout recommendation based on significance, lift, and confidence.
- Developed a Streamlit dashboard and CLI interface to visualize experiment results including conversion rates, confidence intervals, and comparisons between models, giving a user-friendly system for data driven decisions.

ML Debug Assist (Python, scikit-learn, YAML)

- Designed and built a fully local ML-powered debugging assistant that classifies **Python** runtime errors using a hybrid rule-based and machine-learning approach to generate structured troubleshooting guidance.
- Implemented a pipeline covering synthetic dataset generation, text preprocessing, **TF-IDF** feature extraction, and multiclass **Logistic Regression** training.
- Added confidence-aware inference logic to dynamically adjust output, surfacing multiple candidate error categories and prompting for additional context when predictions were uncertain.
- Developed **YAML**-driven fix playbooks with similarity-based retrieval to deliver actionable debugging checklists and contextual examples, enabling reproducible, offline execution via a CLI.

Traffic Sign Image Recognition System (Python, PyTorch, scikit-learn, OpenCV, Pandas, NumPy, Torchvision, Matplotlib)

- Built an image recognition and classification models for **43** traffic sign classes using the GTSRB dataset, training, evaluating, and optimizing multiple machine learning models.
- Developed a **Neural Network** using **ResNet18** and **Transfer Learning** achieving up to a **94%** test accuracy through stage-based training, and fine tuning.
- Implemented multiple baseline models including **HOG + SVC, k-NN, and Random Forest** models, and through hyperparameter tuning was able to improve model test accuracies from as low as **77%** to as high as **97%** improving overall model accuracies by **10%-20%.**
- Evaluated models based statistical analysis using precision, recall, F1-score, and matplot for confusion matrices visualization to analyze class level errors to better tailor optimizations and improve accuracies
- Engineered data preprocessing and augmentation pipelines, including normalization and standardization of the image features.

TECHNICAL SKILLS

- Programming Languages: Python, Java, JavaScript, C++, C, C#, Dart, HTML, CSS, SQL, PHP, YAML
- Technologies: GitHub, Git, Emulators / Simulators, Xcode, Microsoft Office, Google Suite, MySQL, DBeaver, Firebase, Linux, Unix, Command Line, Unit Testing, Data Structures, Algorithms, Flutter
- Core Skills: Object Oriented Programming, Database Management, Full stack Development, App Development, Web Development, Debugging, Troubleshooting, Automation Scripting, UI/UX, Server-Side Programming, Version Control, REST APIs, AI API Integrations, Prompt Engineering, Mobile UI Architecture, Machine Learning Model Development, Data Processing, Data visualization